Contrast Enhancement of Multiple Tissues in MR Brain Images With Reversibility

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***Abstract*—Contrast enhancement (CE) of magnetic resonance (MR) brain images is an important technique to bring out the tissue details for clinical diagnosis. Recently, a new form of image enhancement has been proposed to complete the task without any information loss. Specifically, information required to restore the original image is reversibly hidden into the enhanced image. Moreover, several image segmentation based algorithms have been proposed so that the region of interest can be exclusively enhanced. However, with the reversible algorithms, it is hard to properly enhance the tissues in MR brain images when they are relatively small or connected with each other. To address this issue, a hierarchical CE scheme is proposed for MR brain images with reversibility in this letter. Firstly, a deep convolutional neural network is used to segment multiple tissue classes automatically. Then, the segmented tissues are individually utilized to guide the CE procedure so that individual-tissue-enhanced images are generated. Compared with using the background information to guide the CE procedure, better tissue enhancement effects and visual quality are both obtained by our proposed hierarchical scheme. The evaluation results obtained over MR brain test images demonstrate the reversibility and adaptability of the proposed scheme for the enhancement of interested tissues.**

***Index Terms*—Image segmentation, magnetic resonance image, medical image enhancement, reversible data hiding.**

I. Introduction

MAGNETIC resonance imaging (MRI) plays a key role in clinical diagnosis by using a magnetic field and radio waves to create detailed pictures of the brain or other organs [1], [2]. One advantage of using the MRI is that little harmful radiation is emitted to the bodies, but the resultant image may be with low contrast. Since it is not easy for a clinician to perform diagnosis on the low-contrast images, it will be helpful if the contrast of tissues in magnetic resonance (MR) images can be increased [3]. Thus contrast enhancement (CE) is considered as an important preprocessing in medical image analysis.

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Recently, a new procedure has been proposed to perform the task of image CE without information loss. By using the technique of reversible data hiding (RDH) [4]–[16], all information required to restore the original image is reversibly hidden into the enhanced image [17]–[27]. Additionally, extra data such as patient and authentication information may be hidden in the enhanced image [19], [21], [23], [27], which can be extracted when needed. If a contrast-enhanced image is unsuitable for an application, a new image can be generated after recovering the original one. Compared with the reversible schemes for computed tomography images (e.g., [19], [21], [23]), there are few relevant algorithms for the MR images.

The first work combining image CE with RDH was proposed in [17]. However, directly applying it to medical images does not produce satisfactory effects due to the monotonous background. To improve the visual quality after the CE, the Otsu's method [28] was adopted in [19] for background segmentation in medical images so that the region of interest (ROI) can be specially enhanced. Although better performance was obtained by differentiating the bins in histogram equalization (HE), the scheme proposed in [19] has some limitations. Firstly, the Otsu's method segments an image into two parts according to an automatically calculated threshold. However, the segmentation is not always accurate, especially when the ROI region is small or the image histogram is not subject to the bimodal distribution. In order to improve the accuracy, a user interaction based segmentation algorithm named Grabcut [29] was employed in [27]. As the ROI can be better segmented with the aid of user interactions, difficulties may arise when the ROI regions are connected with each other. Secondly, the algorithms in [19], [27] utilized the segmented background to guide the procedure but they do not work well when the same pixel values are contained in the ROI and the background.

In this letter, we propose a hierarchical CE scheme to improve the performance of pre-screening multiple tissues in the MR images. As quite a few MR image segmentation methods have been proposed (e.g., [30], [31]), a convolutional neural network (CNN) is adopted in our scheme to perform tissue segmentation automatically to enhance multiple tissues individually. Different from the algorithms proposed in [19], [27], the principal grey-level values in each segmented tissue are identified and used to bring out the details. In particular, only the histogram bins of the principal values are labelled and equalized for data embedding. A flowchart of the proposed reversible CE scheme is illustrated in Fig. 1, whereby several tissue-enhanced images may be generated from the same MR image, respectively. The

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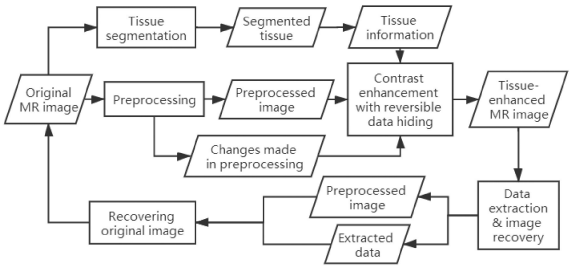

Fig. 1. Flowchart of a scheme for reversible MR image enhancement.

experimental results on a set of MR brain images have shown the efficacy and applicability of our proposed scheme. The evaluation results demonstrate the superior performance in terms of CE effect and image visual quality by comparing with the algorithms in [19] and [27].

In the next section, a procedure of RDH with CE will be introduced, followed by a new scheme for the tissue enhancement of the MR images. The experiments and evaluation results are presented in Section III. Finally, the concluding remarks are drawn in Section IV.

II. Scheme for Reversible MR Image Enhancement

Before proposing a new scheme for the reversible tissue enhancement, a RDH-based procedure will be briefly introduced.

A. RDH-Based Procedure for Image Enhancement

For reversible CE of a grey-level image, a RDH-based method is presented in [17]. Firstly, an image histogram is obtained by counting the number of every grey-level value. In the histogram, the highest two bins are found out and their values are denoted as p_L and p_R ($p_L < p_R$), respectively. By shifting the bins around the chosen bins outward, two empty bins are created so that each of the chosen bins may be expanded into two adjacent bins, respectively. Specifically, the histogram is modified by traversing all pixels in a given order and a pixel value p is changed to p' by

$$p' = \begin{cases} p-1, & \text{if } p < p_L \\ p-b_i, & \text{if } p = p_L \\ p, & \text{if } p_L < p < p_R \\ p+b_i, & \text{if } p = p_R \\ p+1, & \text{if } p > p_R \end{cases} . \quad (1)$$

Note that p is unchanged if it is between p_L and p_R. When $p < p_L$ or $p > p_R$, p is modified but no bit is embedded. For the i-th pixel with value p_L or p_R, the i-th bit value (0 or 1), which is denoted by b_i, is embedded by changing p to p'. After applying Eq. (1) to every pixel that is contained in the histogram, each of the chosen bins are expanded to two adjacent bins. After that, p_L and p_R need to be updated by finding them in the *modified* histogram. By iteratively modifying the histogram by applying Eq. (1), the effects of HE can be achieved while the original pixel values can be restored according to the recorded p_L and p_R values. The bounding pixel values (0 and 255 for a 8-bit pixel value) may be overflowed after histogram modification. To prevent the overflow, a preprocessing needs to be performed to reserve S empty bins on the each side of histogram before expanding S pairs of bins to obtain the effects of HE. In [20], another RDH-based algorithm is proposed by shifting the histogram bins towards one direction at each time. Similarly, the effects of CE can be achieved by iteratively expanding the highest bin in the histogram into two adjacent ones.

Procedure 1: Hierarchical Tissue Enhancement.

Input: MR image I, threshold R and parameter S
Output: Tissue-enhanced image I_c

Segment I into different tissues with the U-Net [32] and use R to identify the principal pixel values in each tissue;
Apply the preprocessing in [24] to I and record the changes (i.e., side information);
Calculate the histogram of the preprocessed image;
foreach *segmented tissue* **do**
  Label the histogram bins of the principal pixel values;
  while *number of expanded bins is less than* $2S-2$ **do**
    Choose the highest two among the labelled bins;
    Expand the chosen bins to embed the side information and the values of the previously expanded bins by applying Eq. (1) to all pixels;
  Collect the LSB values of the last 16 pixels;
  Calculate the histogram excluding the 16 pixels;
  Choose the highest two among the labelled bins;
  Embed the 16 original LSB values, S value and previously expanded bin values by applying Eq. (1);
  Generate the tissue-enhanced image I_c by overwriting the 16 LSB with the last two expanded bin values.

B. Proposed Scheme for Reversible MR Image Enhancement

Based on the aforementioned procedure, a hierarchical CE scheme is proposed for MR images according to the flowchart as shown in Fig. 1. Different from other reversible algorithms, the ROI regions in a MR image are individually utilized to guide the procedure of CE so that several tissue-enhanced images can be generated from one MR image, respectively.

1) Tissue Segmentation: The ROI (e.g., tumors) in MR brain images usually exist in the tissues such as cerebellum and unmyelinated white matter. To recognize and segment different tissues, a CNN-based method is adopted to automatically divide a MR brain image into different areas by their tissue classes. For simplicity of implementation, we use the U-Net [32] as the segmentation module. U-Net is a classic image segmentation framework with 2D CNN, and it works with few training images to yield precise segmentation results.

2) Hierarchical CE of Segmented Tissues: In [19], [27], some bins in the histogram are excluded from being expanded. As the same pixel values may exist in both background and ROI, the interested regions may not be properly brought out. Besides, multiple ROI regions cannot be separately enhanced by applying the methods in [19]. To cope with these issues, a hierarchical CE scheme is proposed in this letter by calculating the percentage of every grey-level value in a segmented tissue. If the percentage of a grey-level value is over a threshold R, the corresponding

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Procedure 2: Recovery of the Original MR Image.

Input: Tissue-enhanced image I_c
Output: Original MR image I

Collect the LSB of the last 16 pixels in I_c to calculate the values of the last expanded bins (i.e., p_{LL} and p_{LR});
Calculate an image histogram excluding the 16 pixels;
Apply Eq. (2) to extract the bit values hidden in the histogram and use Eq. (3) to restore the pixel values;
Obtain the values of the original LSB of the 16 pixels, parameter S and the previously expanded bins from the extracted bit values;
Replace the LSB of the 16 pixels with the original values;
Calculate a new histogram over the whole image;
while *less than S pairs of bin values have been used* **do**
  Apply Eq. (2) to extract the bit values hidden in the histogram and use Eq. (3) to restore the pixel values;
  Obtain the previously expanded bin values from the extracted data and recalculate the histogram;
Obtain the side information from the extracted data and reverse the preprocessing to recover the original image I.

histogram bin is eligible to be expanded for data embedding, while the rest bins can only be shifted in the histogram modification. Since the histogram bins to be equalized are chosen according to each tissue to be enhanced, multiple tissue-enhanced images can be obtained by enhancing every segmented tissue individually.

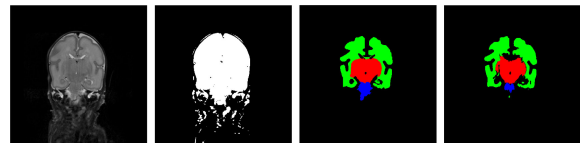
3) Procedure of Hierarchical Tissue Enhancement: The process of tissue CE is presented in Procedure 1 by expanding S pairs of histogram bins so that a tissue-enhanced image I_c is generated from a MR image I. The preprocessing proposed in [24] is adopted, which shrinks the histogram while preserving the order of all pixel values to avoid visual distortions. Note that the least significant bit (LSB) of sixteen pixels are used to save the last two expanded bin values represented in sixteen bits. For simplicity of understanding, the last sixteen pixels are chosen in Procedure 1, whose positions can also be randomly chosen with a secret key. Since the LSB is overwritten only once, no visual pattern will be caused. The value of S needs to be hidden in I_c, while neither the segmentation information nor the value of R is required for original image recovery.

4) Recovery of Original Image: To extract the hidden data, the values of the lastly expanded bins should be known, which are denoted by p_{LL} and p_{LR}, respectively. Then the pixels in a tissue-enhanced image are traversed to find those values equaling to any of $\{p_{LL}-1, p_{LL}, p_{LR}, p_{LR}+1\}$. A bit value denoted by b' may be obtained from a pixel value p' by

$$b' = \begin{cases} 1, & \text{when } p' = p_{LL}-1 \text{ or } p' = p_{LR}+1 \\ 0, & \text{when } p' = p_{LL} \text{ or } p' = p_{LR} \\ \text{null}, & \text{other cases} \end{cases}. \quad (2)$$

Meanwhile, a new pixel value p is generated to replace p' by

$$p = \begin{cases} p'+1, & \text{when } p' < p_{LL} \\ p', & \text{when } p_{LL}-1 < p' < p_{LR}+1 \\ p'-1, & \text{when } p' > p_{LR} \end{cases}. \quad (3)$$



(a) Original (b) Otsu [28] (c) Ground truth (d) U-Net[32]

Fig. 2. MR brain image and segmentation results.

By collecting the extracted bits, the value of S and those of the previously expanded bins can be known. The preprocessed image can be obtained by updating the values of $\{p_{LL}, p_{LR}\}$ in Eq. (2) and Eq. (3) for $S-1$ times. The process of data extraction and image recovery is summarized in Procedure 2.

III. Experimental Results

We applied the proposed scheme and the schemes in [19], [27] to a set of 50 MR brain images, which were downloaded from the NeoBrainS12 challenge website [33]. Each of the test images was associated with a fully annotated ground truth segmentation map of tissue classes.

A. Tissue Segmentation

When applying the proposed scheme, tissue segmentation was conducted by adopting the U-Net [32]. A set of 50 T2-weighted MR images at a resolution of 384 × 384 pixels were used as training data. For illustration, one test MR image and the results obtained with the U-Net [32] and Ostu's method [28] are shown in Fig. 2, respectively. Otsu's method may be applied to separate image background from the foreground, but cannot segment the individual tissue from the MR brain image. By adopting the U-Net, multiple tissues were automatically segmented from the MR images with high accuracy. Similar results were also obtained for other test images.

B. Visual Effects of CE and Reversibility

We compare the visual effects by setting parameters R and S to 0.01 and 40 in the proposed scheme and [19], [27], respectively. In applying the proposed scheme, the grey-level values whose percentages over 1% of the segmented tissue are labelled and the corresponding histogram bins are eligible to be expanded. In applying the schemes in [19], [27], the grey-level values whose percentages over 1% of the background are identified and the corresponding bins are excluded from being expanded in HE. In implementing each scheme, the same number of histogram bins are expanded.

From the MR brain image shown in Fig. 2(a), multiple tissue-enhanced images are obtained with the proposed scheme, as shown in Fig. 3 and Fig. 4. The enhanced tissues are identified according to the ground truth segmentation map, which are zoomed in the red box, respectively. When applying the scheme in [27], tissue segmentation is conducted by manually selecting the background and ROI region. When the regions of the tissues to be segmented are small, the background used to guide the process of CE varies little. The schemes in [19], [27] can only enhance all ROI regions at the same time so that only one contrast-enhanced image is generated from the original MR

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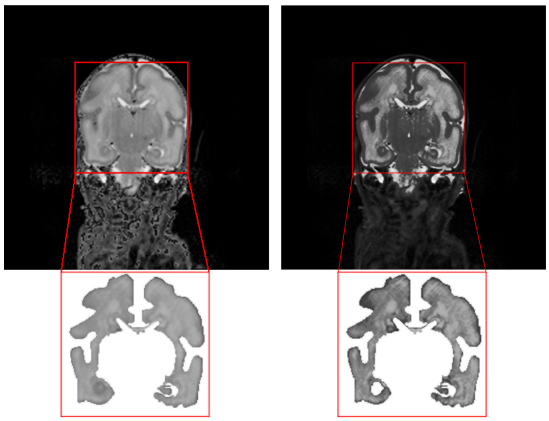
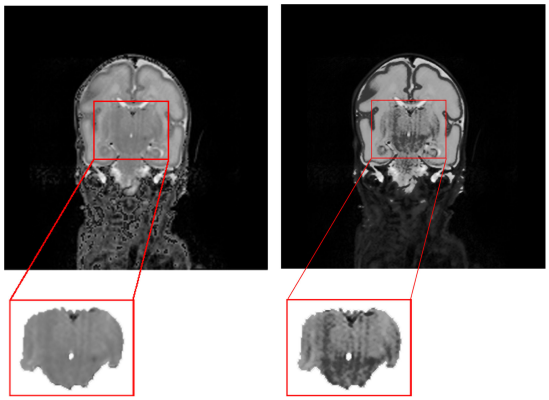


(a) Using the scheme in [19] (b) Using the proposed scheme

Fig. 3. Tissue-enhanced MR images with zoomed tissue ($S = 40$ and $R = 0.01$).



(a) Using the scheme in [19] (b) Using the proposed scheme

Fig. 4. Tissue-enhanced MR images with zoomed tissue ($S = 40$ and $R = 0.01$).

TABLE I
NUMERICAL RESULTS (MEAN) OBTAINED FROM FIVE MR BRAIN IMAGES FOR THE TISSUE CLASS AS SHOWN IN FIG. 3

Evaluation metric	R	2%	1%	0.2%	0.1%	2%	1%	0.2%	0.1%
	Scheme	S=30				S=40			
RCEOI	Proposed	**0.572**	**0.570**	0.562	0.560	**0.601**	**0.590**	0.572	0.570
	[19]	0.510	0.518	0.550	0.551	0.514	0.510	0.563	0.564
	[27]	0.509	0.515	**0.570**	**0.571**	0.516	0.517	**0.587**	**0.597**
REEOI	Proposed	**0.558**	**0.558**	0.554	**0.554**	**0.571**	**0.570**	0.563	0.561
	[19]	0.517	0.527	0.553	0.553	0.522	0.519	0.558	0.559
	[27]	0.521	0.526	**0.557**	0.547	0.527	0.530	**0.569**	**0.566**
RMBEOI	Proposed	**0.983**	**0.982**	0.970	**0.973**	**0.971**	**0.987**	0.938	0.940
	[19]	0.895	0.909	0.952	0.953	0.862	0.882	0.922	0.923
	[27]	0.899	0.907	**0.987**	0.970	0.870	0.876	**0.977**	**0.955**
SSIM	Proposed	**0.961**	**0.961**	**0.964**	**0.965**	**0.941**	0.943	0.947	**0.949**
	[19]	0.896	0.935	0.946	0.945	0.873	0.860	0.911	0.909
	[27]	0.925	0.950	0.962	0.960	0.916	**0.946**	**0.948**	0.941
PSNR	Proposed	**28.23**	**28.39**	**29.14**	**29.36**	**25.48**	**26.13**	26.74	**26.90**
	[19]	23.94	26.01	28.94	28.70	22.04	22.32	25.95	25.84
	[27]	24.49	25.92	28.39	27.99	22.88	23.97	**26.81**	25.40
pure hiding rate (bpp)	Proposed	0.141	0.148	**0.160**	**0.164**	0.108	0.123	0.151	0.157
	[19]	0.254	0.200	0.129	0.128	**0.307**	**0.309**	**0.163**	**0.162**
	[27]	**0.264**	**0.230**	0.138	0.114	0.285	0.252	0.110	0.062

TABLE II
NUMERICAL RESULTS (MEAN) OBTAINED FROM FIVE MR BRAIN IMAGES FOR THE TISSUE CLASS AS SHOWN IN FIG. 4

Evaluation metric	R	2%	1%	0.2%	0.1%	2%	1%	0.2%	0.1%
	Scheme	S=30				S=40			
RCEOI	Proposed	**0.569**	**0.560**	0.548	**0.541**	**0.596**	**0.577**	0.566	0.557
	[19]	0.506	0.507	0.537	0.539	0.510	0.506	0.556	**0.558**
	[27]	0.503	0.507	**0.553**	0.497	0.506	0.507	**0.568**	0.495
REEOI	Proposed	**0.549**	**0.544**	**0.540**	**0.537**	**0.564**	**0.560**	**0.552**	**0.547**
	[19]	0.514	0.515	0.534	0.534	0.518	0.512	0.541	0.541
	[27]	0.517	0.520	0.534	0.499	0.517	0.517	0.547	0.492
RMBEOI	Proposed	**0.985**	**0.989**	**0.993**	**0.990**	**0.972**	**0.972**	**0.987**	**0.990**
	[19]	0.911	0.933	0.958	0.954	0.884	0.897	0.957	0.953
	[27]	0.914	0.929	0.971	0.942	0.892	0.900	0.975	0.912
SSIM	Proposed	**0.961**	**0.963**	0.966	0.968	**0.943**	**0.948**	0.952	0.955
	[19]	0.896	0.935	0.946	0.945	0.873	0.860	0.911	0.909
	[27]	0.925	0.951	**0.969**	**0.991**	0.916	0.946	**0.958**	**0.982**
PSNR	Proposed	**27.71**	**28.09**	28.79	29.41	**25.06**	**25.54**	26.14	26.61
	[19]	23.94	26.01	**28.94**	28.70	22.04	22.32	25.95	25.84
	[27]	24.49	26.03	28.47	**31.97**	22.88	24.07	**26.46**	**28.40**
pure hiding rate (bpp)	Proposed	0.117	0.127	**0.140**	**0.148**	0.088	0.114	0.137	0.145
	[19]	0.254	0.200	0.129	0.128	**0.307**	**0.309**	**0.163**	**0.162**
	[27]	**0.264**	**0.233**	0.084	0.001	0.285	0.254	0.037	0.001

image. As shown in Fig. 3 and Fig. 4, each tissue in the original MR brain image can be adaptively enhanced with the proposed scheme because the details of each ROI are better brought out. With all of the schemes compared, each original MR image was correctly recovered from its enhanced versions.

C. Numerical Evaluation of Tissue Enhancement

To measure the effects of CE, relative entropy error, relative contrast error and relative mean brightness error defined in [34] are calculated in each tissue by referring to the ground truth, respectively denoted by RCEOI, REEOI and RMBEOI. The peaked signal-to-noise ratio (PSNR) and Structural SIMilarity (SSIM) index [35] are also calculated between the original and contrast-enhanced images. By applying the proposed scheme and those in [19], [27], the average results obtained from five MR brain images are listed in Table I and Table II for different values of S and R. The data hiding rate is defined as the number of extra bits hidden per pixel (bpp in short) so that reversibility is achieved with any positive rate.

It can be seen that better CE effects are achieved with the proposed scheme because the largest REEOI and RCEOI are obtained for $R = 1\%$ and $R = 2\%$, respectively. The scheme in [27] performs slightly better with a smaller R, while less CE effects are achieved with the scheme in [19]. In general, the SSIM, RMBEOI and PSNR are better at a lower value of S whereas both contrast (RCEOI) and entropy (REEOI) benefit from larger S. When applying the schemes in [19], [27], the resultant effects vary a lot along with R. The performance of the proposed scheme is consistently good, indicating that better adaptability for the tissue CE in the MR images is achieved.

IV. CONCLUSION

We have proposed a hierarchical tissue enhancement scheme for MR images by adopting a preprocessing strategy and employing the U-Net [32] for segmentation. By analyzing each segmented tissue, different tissues in MR brain images can be individually enhanced to bring out the details. The numerical results on a set of MR images have shown the reversibility and better adaptability of the proposed scheme. In general, there is a tradeoff between tissue contrast enhancement and visual quality preservation when applying the reversible methods.

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